Akoizo Original Practice - Form 3 - Answer key

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A 5.0 cm wheel turns 40 times without slipping. About how far does the vehicle travel?

A. 6.28 m

B. 2.00 m

C. 12.57 m

D. 0.63 m

Correct: A

2. (2 points) The vehicle stops 8.0 cm from the Target Point. How many Distance Score points does that add?

A. 16.0

B. 8.0

C. 4.0

D. 32.0

Correct: A

3. (2 points) Which energy source is allowed?

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

A. A wound spring to assist launch

B. A large capacitor bank

C. Eight labeled 1.2 V NiMH AA cells

D. A lithium-ion pack

Correct: C

4. (2 points) Which tool use is prohibited during event time?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. Adjusting the program over Bluetooth

B. Using a tape measure to check the track

C. Using AI to adjust the program

D. Programming over USB

Correct: C

5. (4 points) Write the formula for the reverse distance in terms of MP forward travel and Target Distance.

answer: Reverse distance = MP forward travel - Target Distance. (4 points).

6. (4 points) List three items to impound.

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

answer: The vehicle, its batteries stored separately, and spare parts; also any alignment device, papers, and the Kit Modification Form if used. (4 points).

7. (4 points) Describe how a V-shaped entry helps capture the bottle.

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

answer: Angled guides contact an off-center bottle and slide it toward the middle pocket as the vehicle moves forward. (4 points).

8. (4 points) How can two encoders keep a vehicle straight?

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

answer: The controller compares left and right counts and adjusts motor speeds to keep them equal. (4 points).

9. (4 points) What is the Final Score if run 1 scores 118.17 and run 2 scores 109.79?

answer: 109.79, the better (lower) of the two runs. (4 points).

10. (4 points) Why must all energy sources be easy to access?

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

answer: So the supervisor can inspect batteries and other sources to verify they are legal. (4 points).

11. (6 points) Solve this case and explain your reasoning: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?

reasoning: Time outside cruise: 4 x 1.0 + 0.5 + 0.3 = 4.8 s; cruise time = 10.2 s.
Each ramp covers about half the cruise speed for 1 s, so four ramps about 2 x v.
Distance: 10.2v + 2v = 10.20 m, so v about 0.84 m/s. (3 points).

conclusion: Cruise at about 0.84 m/s, then fine-tune from measured run times. (3 points).

12. (2 points) What went wrong in the plan?

Supplied synthetic observations:
Setup: Target 8.00 m, Bottle Line 1.00 m, vehicle length 45 cm.
Run: MP forward travel programmed as 8.60 m.
Result: Bottle ended 1 cm short of the line.

A. The reverse was too long

B. The Target Time was wrong

C. The forward travel ignored the bottle diameter and margin

Correct: C

13. (2 points) What is the team’s Final Score?

Supplied synthetic observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. The average of the two runs

B. 96.4

C. 2500

Correct: B

14. (2 points) Define “Bottle Pusher” in the context of Electric Vehicle.

answer: Rigid front device that catches and pushes the bottle. (2 points).

15. (2 points) Define “Rotary encoder” in the context of Electric Vehicle.

answer: Sensor that outputs pulses as a shaft turns. (2 points).

16. (2 points) Define “Bottle Distance” in the context of Electric Vehicle.

answer: Distance from the Bottle Line to the bottle’s closest side. (2 points).

17. (2 points) Define “Kit Modification Form” in the context of Electric Vehicle.

answer: Document of two functional changes to a kit; impounded. (2 points).

18. (2 points) Define “Trapezoidal profile” in the context of Electric Vehicle.

answer: Accelerate, cruise, decelerate motion plan. (2 points).